Benyamin Delshad



Backend Software Engineer (Kotlin · Java)

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Backend engineer with **4+ years of professional Java and Kotlin experience** building microservices with **Spring Boot, Kafka, and event-driven architecture**, including distributed workflow orchestration with Temporal, taken from an empty repository to production, guided by Domain-Driven Design, TDD, and continuous delivery. Strong computer-science foundation: research background in distributed systems and algorithms, top-10 rank at ACM-ICPC Asia Region. Industry experience across **e-commerce and ticketing, energy, and logistics**.

CORE SKILLS

Languages	Kotlin, Java, Python, SQL, Bash, C/C++
Backend	Spring Boot, REST APIs, microservices, event-driven architecture, Apache Kafka, Avro + Confluent Schema Registry, Temporal workflow orchestration, idempotency, dead-letter queues
Data	PostgreSQL, Oracle DB, Flyway migrations, JPA/Hibernate
Infra & Ops	Kubernetes, Helm, ArgoCD, Docker/Podman, GitHub Actions, GitLab CI/CD, AWS S3, Confluent Cloud, Micrometer and structured logging, k6 load testing, JVM memory tuning
Practices	Domain-Driven Design, hexagonal architecture, Clean Code, TDD (JUnit, Kotest, MockK, Testcontainers), Temporal replay tests, SonarCloud quality gates, Scrum, pair programming, code review

WORK EXPERIENCE

Software Engineer, Platform Integration (Yatta Checkout product)

07/2026 – present

Yatta Solutions GmbH · Frankfurt am Main. Yatta Checkout is Yatta's own e-commerce platform for digital goods, covering payment, fulfillment, and delivery.

- Architected and built a greenfield, **event-driven fulfillment service** (Kotlin, Spring Boot, Temporal) that turns a completed purchase into tickets, wallet links, a delivery email and a print-at-home PDF. Live since September 2026 for a regional public-transport association: **about 10,000 production workflows in its first ten days with zero failed**.
- **Load-tested with k6** against real webhooks before go-live and found a concurrency race in the Temporal workflow that dropped early Kafka replies. Fixed it replay-safely with workflow versioning and replay tests: **clean runs from 79% to 100%** under load, worst case from ~5 minutes to 34 s. The first overnight batch then ran **2,532 subscription renewals with zero failures** (p95 2.1 s).
- Diagnosed a **JVM Metaspace out-of-memory crash** under load on Kubernetes and fixed the buildpack's memory sizing, taking restarts under load to zero and unblocking the production rollout.
- **Ran the production releases** (GitOps with Helm and ArgoCD), set the release order across dependent services, and kept go-live checklists with rollback plans. After a deployment-downtime incident, filed the analysis and proposed splitting the HTTP API from the workflow worker.
- Built **Kafka and Avro consumer pipelines** with per-listener retry classification, a dead-letter path and stalled-consumer metrics, and evolved event contracts across services without breaking consumers or in-flight workflows.
- Implemented **VDV-KA compliant Aztec barcode rendering** (ZXing plus a hand-written 1-bit PNG encoder), verified by decode-back tests; the barcodes are scanned by real ticket-inspection devices.
- Structured the service as **independently extractable bounded contexts** (DDD, hexagonal), with streaming document paths and sanitized error boundaries that keep signed URLs and personal data out of logs. Reviewed colleagues' pull requests; reviews surfaced lost-update races and duplicate-email risks and changed final designs.

Stack: Kotlin, Spring Boot, Temporal, Apache Kafka, Avro, PostgreSQL, Flyway, AWS S3, Kubernetes, Helm, ArgoCD, k6, Gradle

Software Engineer, SMA Solar Technology project (energy domain) 12/2023 – 06/2026

Yatta Solutions GmbH (consulting division) · Frankfurt am Main. Client project for SMA, a leading solar-inverter manufacturer.

- Developed an **EEBUS protocol driver in Java** to process grid-based power-limitation requests, ensuring compliance with German energy regulations (§ 14a EnWG, § 9 EEG).
- Built core **energy-management services and APIs** in a cross-functional team, optimizing end-user energy consumption and costs.
- Architected **GitLab CI/CD pipelines** automating builds, code formatting, static analysis, and test coverage.
- Sustained **80%+ customer satisfaction (NPS)** over two and a half years of iterative delivery in a Scrum team.

Stack: Java, REST APIs, Python, Bash, TypeScript, GitLab CI

Software Engineer, Rhenus Logistics project (logistics domain) 04/2022 – 11/2023

Yatta Solutions GmbH (consulting division) · Frankfurt am Main. Client project for Rhenus, one of Europe's largest logistics providers.

- Cut shipment-processing overhead with an automated **compliance system that proactively blocks high-risk shipments** to sensitive destinations.
- Engineered a **Spring Boot REST service providing end users with real-time shipment quotes**.
- Achieved **80%+ customer satisfaction (NPS)**.

Stack: Java, Spring Boot, REST APIs, JavaScript, Oracle DB

Doctoral Researcher & Teaching Assistant 10/2020 – 03/2022

University of St. Gallen · Switzerland

- Accelerated distributed-systems development by designing a **domain-specific language that automates operation-level consistency inference** (Java, NoSQL, JML).
- Taught in courses on engineering software systems, software engineering, and programming for data science.

Research Intern (M.Sc. thesis), Game Theory & Simulation 02/2020 – 09/2020

Institute of Science and Technology Austria (ISTA) · Klosterneuburg. Supervised by Prof. Krishnendu Chatterjee.

- Validated theoretical research by building **evolutionary game-theory simulations** to analyze probabilistic cooperation strategies (MATLAB).

Research Intern, Graph Algorithms for Transit Routing 04/2019 – 09/2019

IRIF (Institute on the Foundations of Computer Science) · Paris

- Modeled **public-transit networks as temporal graphs** and designed algorithms to assess infrastructure changes using real-world datasets (C/C++).

EDUCATION

M.Sc. Computer Science, Université Paris Cité (MPRI) 09/2018 – 09/2020

Paris, France. Merit-based scholarship; research-track master's (Master Parisien de Recherche en Informatique).

B.Sc. Computer Science, University of Tehran 09/2014 – 08/2018

Ranked 8th, ACM-ICPC Asia Region (2016) · two-time bronze medalist, Iranian National Mathematics Competition (2016, 2017) · teaching assistant for algorithms, graph theory, and programming courses.

LANGUAGES

English	Fluent (C1)
German	Intermediate (B1), currently taking a course
Persian	Native

INTERESTS

Travel and local cuisines, Middle-earth linguistics, Wizarding World lore, detailed LEGO models.